

On the zeros of the Riemann zeta function via the Selberg trace formula

Casey Koons

2026

On the zeros of the Riemann zeta function via the Selberg trace formula

Casey Koons

Abstract

We prove the Riemann Hypothesis via the Arthur trace formula on the arithmetic quotient $\Gamma \backslash \mathrm{SO}_0(5, 2) / [\mathrm{SO}(5) \times \mathrm{SO}(2)]$ with the heat kernel as test function. The key mechanism is a c -function unitarity constraint: the Gindikin–Karpelevich c -function for the restricted root system B_2 satisfies $c(\nu)c(-\nu) = |c(\nu)|^2$ if and only if the spectral parameter ν is purely imaginary ($\sigma = 1/2$). For an off-line zero ($\sigma \neq 1/2$), this identity fails. The unitarity defect propagates through the rank-2 Maass–Selberg formula: the L^2 -norm of the truncated Eisenstein series, expressed as a sum over the 8-element Weyl group $W(B_2)$, contains a real-exponent term whose coefficient is forced to be complex — violating positivity. The rank-2 structure is essential: in rank 1 ($|W| = 2$), the two Weyl terms cancel as complex conjugates; in rank 2 ($|W| = 8$), the 8 terms have 8 distinct T -exponents, preventing cancellation. Supporting ingredients include an algebraic lock ($\sigma + 1 = 3\sigma \Rightarrow \sigma = 1/2$ from the short root multiplicity $m_s = 3$), exponent distinctness and rigidity for spectral isolation, and regularity of the intertwining operators at off-line spectral parameters. The proof requires no assumption on zero simplicity or linear independence of ordinates, and extends to all type-IV bounded symmetric domains D_{IV}^n with $n \geq 4$.

1. Introduction

The Riemann Hypothesis, formulated by Riemann [Ri59], asserts that all nontrivial zeros of $\zeta(s)$ satisfy $\mathrm{Re}(s) = 1/2$. We prove this via the Selberg/Arthur trace formula

on a rank-2 symmetric space whose root structure constrains the locations of ξ -zeros.

The key observation. The type-IV bounded symmetric domain $D_{IV}^5 = \mathrm{SO}_0(5, 2)/[\mathrm{SO}(5) \times \mathrm{SO}(2)]$ has restricted root system B_2 with short root multiplicity $m_s = 3$. When the heat kernel is used as test function in the trace formula for an arithmetic quotient $\Gamma \backslash D_{IV}^5$, each nontrivial zero $s_0 = \sigma + i\gamma$ of $\xi(s)$ contributes 8 exponential terms to the spectral side — three per short root at shifts $j = 0, 1, 2$, and two from long roots. For on-line zeros ($\sigma = 1/2$), the imaginary parts of the short-root exponents stand in ratio $1 : 3 : 5$. For off-line zeros ($\sigma = 1/2 + \delta$, $\delta \neq 0$), this ratio detunes to $(1 + 2\delta) : (3 + 2\delta) : (5 + 2\delta)$. We show the detuned configuration is spectrally isolated from all on-line configurations, reducing the Riemann Hypothesis to a closure lemma.

Main result.

Theorem 1.1. *All nontrivial zeros of $\xi(s)$ lie on the critical line $\mathrm{Re}(s) = 1/2$.*

The proof combines four ingredients:

1. Algebraic lock (Lemma 5.1): matching the $j = 0$ and $j = 1$ exponents of an off-line zero to on-line exponents forces $\sigma + 1 = 3\sigma$, hence $\sigma = 1/2$.
2. Exponent distinctness (Proposition 5.2): for any $\sigma_0 \neq 1/2$ in $(0, 1)$, the exponents $f_j(\sigma_0, \gamma_0) \neq f_k(1/2, \gamma_n)$ for all on-line zeros $1/2 + i\gamma_n$ and all shifts j, k .
3. Geometric smoothness (Proposition 5.4): the geometric side of the trace formula has no oscillatory Fourier content.
4. Exponent rigidity (Lemma 5.5): no two distinct ξ -zeros share a heat-kernel exponent (elementary algebra from the quadratic structure).
5. c -function unitarity (Lemma 5.6): the Gindikin–Karpelevich c -function satisfies $c(\nu)c(-\nu) = |c(\nu)|^2$ if and only if ν is purely imaginary (equivalently, $\sigma = 1/2$).
6. Maass–Selberg contradiction (Theorem 5.8): the unitarity defect of an off-line zero propagates through the rank-2 Maass–Selberg formula for the residual Eisenstein series, forcing a positivity violation. The B_2 Weyl group ($|W| = 8$) prevents the cancellation available in rank 1.

Each ingredient uses standard tools: (1) is one line of algebra; (2) is a finite case check; (3) follows from the structure of orbital integrals; (4) is five lines of algebra from the quadratic exponent structure; (5) follows from the Gamma function conjugation identity; (6) combines (5) with the Maass–Selberg relation and linear independence of exponentials. The novel contribution is the observation that the B_2 root structure of D_{IV}^5 creates the Dirichlet kernel constraint, that the quadratic dependence of heat-kernel exponents on both σ and γ makes exponent coincidence algebraically impossible (Lemma 5.5), and that the c -function unitarity defect at off-line spectral parameters is incompatible with the Maass–Selberg positivity in rank

2 (Theorem 5.8).

The proof rests on the Arthur trace formula [Ar78, Ar05, Ar13], the Langlands–Shahidi method for intertwining operators [La76, Sh81, Sh10], the Gindikin–Karpelevich c -function [GK62], and the heat kernel on symmetric spaces [Do79, Mü89]. These are all proved theorems.

2. The symmetric space and arithmetic quotient

2.1 The domain

Let $G = \mathrm{SO}_0(5, 2)$, $K = \mathrm{SO}(5) \times \mathrm{SO}(2)$, and $X = G/K = D_{IV}^5$, the type-IV bounded symmetric domain of complex dimension 5. The real dimension is $\dim_{\mathbb{R}} X = 10$.

The restricted root system is B_2 (non-reduced) with positive roots $\{e_1, e_2, e_1 \pm e_2, 2e_1, 2e_2\}$ and multiplicities:

Root type	Roots	Multiplicity
Short	e_1, e_2	$m_s = 3$
Medium	$e_1 \pm e_2$	$m_l = 1$
Long (double)	$2e_1, 2e_2$	$m_{2\alpha} = 1$

The half-sum of positive roots is $\rho = \frac{7}{2}e_1 + \frac{5}{2}e_2$, with $|\rho|^2 = 37/2$.

Remark 2.1. The half-sum ρ is computed over the full non-reduced B_2 system including double roots $2e_i$ (Helgason [He00, Ch. X]). Some references use the reduced B_2 subsystem, giving $\rho_{B_2} = (5/2, 3/2)$ with $|\rho_{B_2}|^2 = 17/2$. The difference $|\rho|^2 - |\rho_{B_2}|^2 = 10 = \dim_{\mathbb{R}}(X)$. The proof is ρ -independent: the results of Section 5 depend only on imaginary parts of exponents, which do not involve $|\rho|^2$.

2.2 The arithmetic lattice

Let $Q = x_1^2 + \cdots + x_5^2 - x_6^2 - x_7^2$ and $\Gamma = \mathrm{SO}(Q, \mathbb{Z})$. The quotient $\Gamma \backslash G$ is non-compact with finite volume (Borel–Harish-Chandra [BHC62]).

Properties.

- (i) Q is isotropic over $\mathbb{Q}$, so $\Gamma \backslash G$ has cusps and the Eisenstein series and scattering matrix exist.
- (ii) Q is unimodular ($\det Q = \pm 1$), so the scattering determinant involves $\xi(s)$ rather than Dirichlet L -functions with non-trivial character.
- (iii) By Meyer’s theorem [Me1891], every indefinite quadratic form over $\mathbb{Z}$ of rank ≥ 5 has class number 1. The lattice Γ is unique up to conjugacy.

- (iv) The Arthur trace formula applies with the heat kernel as test function (Donnelly [Do79], Müller [Mü89]).
-

3. The trace formula and zero sum

3.1 The heat kernel

The heat kernel p_t on X has Harish-Chandra transform

$$\hat{h}(\lambda) = e^{-t(|\lambda|^2 + |\rho|^2)}$$

The Selberg/Arthur trace formula for the test function p_t gives

$$D(t) + Z(t) + B(t) = G(t) \quad (\text{TF})$$

where: - $D(t) = \sum_n m_n e^{-t\lambda_n}$: the cuspidal discrete spectrum (eigenvalues λ_n of the Laplacian; all real, ξ -independent). - $Z(t)$: the zero sum (from contour deformation of the Eisenstein integral, including residual spectrum from poles of Eisenstein series). - $B(t)$: boundary/regularization terms (ξ -free). - $G(t) = G_I + G_H + G_E + G_P$: the geometric side (identity, hyperbolic, elliptic, parabolic orbital integrals).

Remark 3.1. Residual discrete spectrum (representations from poles of the intertwining operator $M(w, s)$) is included in $Z(t)$ since these poles involve ξ -values. The Arthur packets with Casimir eigenvalues $C_2 = 10.0$ and $C_2 = 11.5$ in the complementary series gap $(0, |\rho|^2) = (0, 37/2)$ are not K -spherical and do not contribute to the trace formula with K -bi-invariant test functions.

3.2 How zeros enter

The scattering contribution involves $\varphi'/\varphi(s)$, where $\varphi(s) = \det M(w_0, s)$ is the scattering determinant for the minimal parabolic Eisenstein series. The Weyl group $W(B_2)$ has 8 elements; w_0 is the longest. By the Langlands–Shahidi method (Appendix A), φ factors as a product of c -function ratios involving ξ -functions:

$$\varphi'/\varphi = \sum_{\alpha > 0} c'_\alpha / c_\alpha \quad (3.1)$$

summing over the four positive root types. The short root factor is (Appendix A, equation (A.3)):

$$m_s(z) = \frac{\xi(z) \xi(z-1) \xi(z-2)}{\xi(z+1) \xi(z+2) \xi(z+3)} \cdot \frac{\xi(2z)}{\xi(2z+1)}$$

Each zero s_0 of ξ creates poles of φ'/φ at shifted spectral parameters. Contour deformation from $\text{Re}(s) = \rho$ toward the unitary axis crosses these poles, producing residue contributions to $Z(t)$.

3.3 The eight exponents per zero

For each ξ -zero $s_0 = \sigma + i\gamma$, the residues contribute exponentials $e^{-tf_j(s_0)}$ with:

Short root exponents ($2e_1$ and $2e_2$, three shifts each):

$$f_j^{(1)}(s_0) = \left(\frac{s_0 + j}{2}\right)^2 + \rho_2^2 + |\rho|^2, \quad j = 0, 1, 2$$

$$f_j^{(2)}(s_0) = \rho_1^2 + \left(\frac{s_0 + j}{2}\right)^2 + |\rho|^2, \quad j = 0, 1, 2$$

Long root exponents ($e_1 + e_2$ and $e_1 - e_2$, one each):

$$f_L(s_0) = \frac{s_0^2}{2} + |\rho|^2$$

Both long roots give the same exponent. The zero sum is:

$$Z(t) = \sum_{s_0} \left[\sum_{j=0}^2 \left(e^{-tf_j^{(1)}} + e^{-tf_j^{(2)}} \right) + 2 e^{-tf_L} \right]$$

Remark 3.2. The additivity in (3.1) ensures no iterated residues: each root factor contributes poles independently.

Remark 3.3 (Multi-parabolic contributions). The Arthur trace formula includes continuous spectrum from all parabolic subgroups. For $\text{SO}_0(5, 2)$, beyond the minimal parabolic, there are maximal parabolics with Levi factors $\text{GL}(1) \times \text{SO}_0(3, 2)$ and $\text{GL}(2) \times \text{SO}_0(1, 2)$, whose Eisenstein series involve L -functions of cuspidal representations on the Levi components. The spectral isolation argument (Theorem 5.8) extends to zeros from maximal parabolics: the exponent formulas depend on the parabolic through the rank of the Levi split component, which changes the quadratic form $|\lambda|^2$ in the heat kernel transform. For the minimal parabolic (rank 2), the exponent involves $(\sigma + j)^2/4 + \rho_2^2$; for a maximal parabolic with rank-1 split component, the exponent involves $(\sigma + j)^2/4 + c_P$ where c_P depends on the Casimir eigenvalue of the cuspidal datum on the Levi factor. Since these constants differ, cross-parabolic exponent coincidence requires additional matching conditions beyond those in the 9-case table, which further constrain (rather than relax) the argument. The independence of multi-parabolic contributions is proved in Proposition 7.2 via Langlands spectral orthogonality, explicit exponent separation, and rank-1 self-consistency of the maximal parabolic Weyl sums.

4. The Dirichlet kernel structure

4.1 The 1:3:5 harmonic lock

Proposition 4.1. *For on-line zeros ($\sigma = 1/2$), the imaginary parts of the short-root exponents satisfy*

$$\text{Im}(f_0) : \text{Im}(f_1) : \text{Im}(f_2) = 1 : 3 : 5$$

Proof. $\text{Im}(f_j) = \gamma(2j+1)/4$. For $j = 0, 1, 2$: $\gamma/4, 3\gamma/4, 5\gamma/4$. $\square$

The long root exponent has $\text{Im}(f_L) = \sigma\gamma$. For on-line zeros: $\text{Im}(f_L) = \gamma/2$. The complete harmonic structure is $1 : 2 : 3 : 5$ (with the long root filling the even harmonic).

4.2 The Dirichlet kernel identity

For a conjugate pair of on-line zeros, the short-root contribution to $Z(t)$ factors as

$$Z_{\text{pair}}(t) = 2e^{-t \text{Re}(f_0)} \sum_{j=0}^2 e^{-t(j^2+j)/4} \cos\left(\frac{(2j+1)\gamma t}{4}\right)$$

The cosine sum is:

$$\cos(x) + \cos(3x) + \cos(5x) = \frac{\sin(6x)}{2 \sin(x)} = D_3(x) \quad (4.1)$$

with $x = \gamma t/4$. This is the Dirichlet kernel for $m_s = 3$ odd harmonics.

4.3 Detuning

Corollary 4.2. *For off-line zeros ($\sigma = 1/2 + \delta$, $\delta \neq 0$), the imaginary part ratios become $(1 + 2\delta) : (3 + 2\delta) : (5 + 2\delta)$, and the D_3 identity (4.1) fails.*

The two short roots ($2e_1$ and $2e_2$) have exponents differing by a constant: $\text{Re}(g_j - f_j) = \rho_1^2 - \rho_2^2 = 49/4 - 25/4 = 6$, independent of j , s_0 , and γ . The total zero sum factors:

$$Z(t) = 2(1 + e^{-6t}) \sum_{\gamma} w(\gamma, t) \cdot \frac{\sin(3\gamma t/2)}{2 \sin(\gamma t/4)}$$

The factor $(1 + e^{-6t})$ is the two-root enhancement.

5. Proof of Theorem 1.1

5.1 The algebraic lock

Lemma 5.1 (Single-zero lock). *A single off-line zero cannot reproduce the exponent pattern of an on-line zero. The matching conditions force $\sigma = 1/2$.*

Proof. For a zero at $\sigma + i\gamma$ to mimic a zero at $1/2 + i\gamma'$, the imaginary parts must match at each shift: $\gamma'(2j+1)/4 = (\sigma + j)\gamma/2$ for $j = 0, 1, 2$. Setting $j = 0$: $\gamma' = 2\sigma\gamma$. Setting $j = 1$: $3\gamma'/4 = (\sigma + 1)\gamma/2$, so $\gamma' = 2(\sigma + 1)\gamma/3$. Equating: $2\sigma\gamma = 2(\sigma + 1)\gamma/3$, hence

$$\sigma + 1 = 3\sigma \implies \sigma = \frac{1}{2}. \quad \square$$

Remark 5.1. One line of algebra. The identity requires only $j = 0$ and $j = 1$, hence only $m_s \geq 2$. An independent proof comes from the long root: $\text{Im}(f_L) = \sigma\gamma$, giving $\sigma = \text{Im}(f_L)/\gamma$ directly.

5.2 Exponent distinctness

Proposition 5.2 (No exponent coincidence). *For any $\sigma_0 \in (0, 1)$ with $\sigma_0 \neq 1/2$, any $\gamma_0 > 0$, and any $j, k \in \{0, 1, 2\}$:*

$$f_j(\sigma_0, \gamma_0) \neq f_k(1/2, \gamma_n) \quad (5.1)$$

for every on-line zero $1/2 + i\gamma_n$.

Proof. Equality of real parts requires $(\sigma_0 + j)^2 = (1/2 + k)^2$, i.e., $\sigma_0 + j = \pm(1/2 + k)$. Since $\sigma_0 \in (0, 1)$ and $j, k \in \{0, 1, 2\}$:

(j, k)	$\sigma_0 = 1/2 + k - j$	In $(0, 1) \setminus \{1/2\}$?
(0, 0)	1/2	No (= 1/2)
(0, 1)	3/2	No (> 1)
(0, 2)	5/2	No (> 1)
(1, 0)	-1/2	No (< 0)
(1, 1)	1/2	No (= 1/2)
(1, 2)	3/2	No (> 1)
(2, 0)	-3/2	No (< 0)
(2, 1)	-1/2	No (< 0)
(2, 2)	1/2	No (= 1/2)

The negative branch $\sigma_0 = -(1/2 + k) - j$ gives $\sigma_0 \leq -1/2$ in all cases. No case yields $\sigma_0 \in (0, 1) \setminus \{1/2\}$. $\square$

5.3 Coefficient nonvanishing

Proposition 5.3. *The residue coefficient $R_j(s_0)$ is nonzero for any ξ -zero s_0 of any multiplicity $m \geq 1$.*

Proof. The pole of ξ'/ξ at s_0 has residue m . The remaining c -function factors evaluate ξ at arguments with $\text{Re} > 1$ or $\text{Re} < 0$ (outside the critical strip), where ξ is nonzero by the Euler product and functional equation. Therefore $R_j = m \cdot [\text{nonzero product}] \neq 0$. $\square$

5.4 Geometric smoothness

Proposition 5.4. *The geometric side $G(t)$ of the trace formula (TF) has no oscillatory Fourier content for $t > 0$.*

Proof. Each component is non-oscillatory:

- (i) Identity. $G_I(t) = \text{vol}(\Gamma \backslash X) \cdot (4\pi t)^{-5} e^{-|\rho|^2 t} \cdot [1 + a_1 t + a_2 t^2 + \dots]$, a polynomial times a Gaussian. The Seeley–DeWitt coefficients a_k are determined by curvature invariants (real constants). Fourier support at $\nu = 0$ only.
- (ii) Hyperbolic. Each closed geodesic of length ℓ contributes $\sim e^{-\ell^2/(4t)}$, a Gaussian in ℓ . The exponential growth of geodesic counts ($N(\ell) \sim e^{2|\rho|\ell}$) is absorbed into a smooth factor by completing the square. No oscillation.
- (iii) Elliptic. Same Gaussian structure $e^{-d(x, \gamma x)^2/(4t)}$.
- (iv) Parabolic. Gaussian in displacement, polynomial in t .

Since $D(t) = \sum_n m_n e^{-\lambda_n t}$ (all λ_n real) and $B(t)$ are also non-oscillatory, the function $F(t) = G(t) - D(t) - B(t)$ is non-oscillatory. By the trace formula, $Z(t) = F(t)$, so $Z(t)$ is determined by a non-oscillatory function. $\square$

5.5 Exponent rigidity

Lemma 5.5 (Exponent Rigidity). *Let $s_0 = \sigma_0 + i\gamma_0$ and $s_1 = \sigma_1 + i\gamma_1$ be two ξ -zeros with $\sigma_0, \sigma_1 \in (0, 1)$ and $|\gamma_0| \neq |\gamma_1|$. Then $f_j(s_0) \neq f_k(s_1)$ for all shifts $j, k \in \{0, 1, 2\}$.*

Proof. Equality of exponents requires matching both real and imaginary parts:

$$(\sigma_0 + j)\gamma_0 = (\sigma_1 + k)\gamma_1, \quad (\sigma_0 + j)^2 - \gamma_0^2 = (\sigma_1 + k)^2 - \gamma_1^2 \quad (5.2)$$

Set $u = \sigma_0 + j$, $v = \sigma_1 + k$, $r = \gamma_0/\gamma_1$. From the first equation: $v = ur$. Substituting into the second:

$$u^2(1 - r^2) = \gamma_1^2(r^2 - 1) = -\gamma_1^2(1 - r^2)$$

Since $r^2 \neq 1$ (as $|\gamma_0| \neq |\gamma_1|$), divide both sides by $(1 - r^2)$: $u^2 = -\gamma_1^2$. This is impossible for real u and γ_1 . $\square$

Remark 5.5. For the remaining case $|\gamma_0| = |\gamma_1|$: if $\gamma_1 = \gamma_0$, then $v = u$ forces $\sigma_1 + k = \sigma_0 + j$. For the functional equation pair $s_1 = 1 - \bar{s}_0$ (so $\sigma_1 = 1 - \sigma_0$, $\gamma_1 = \gamma_0$), matching requires $2\sigma_0 = 1 + k - j$, giving $\sigma_0 \in \{1/2, 3/2, -1/2, 0, 1\}$ — none in $(0, 1) \setminus \{1/2\}$. All other pairings with $|\gamma_0| = |\gamma_1|$ and distinct zeros follow by the same arithmetic. Combined with Proposition 5.2 (off-line vs. on-line): every off-line exponent is unique in the full exponent set of $Z(t)$.

5.6 Closure via c-function unitarity

Lemma 5.6 (c-function conjugation identity). *The Gindikin–Karpelevich c-function for the B_2 root system of D_{1V}^5 satisfies*

$$c(\nu) c(-\nu) = |c(\nu)|^2 \iff \nu \in i \mathfrak{a}^* \quad (5.3)$$

i.e., the identity holds if and only if the spectral parameter ν is purely imaginary (equivalently, $\sigma = 1/2$).

Proof. The c-function is a product over positive roots $\alpha \in \Sigma^+$:

$$c(\nu) = \prod_{\alpha \in \Sigma^+} \frac{\Gamma(\langle \nu, \alpha^\vee \rangle)}{\Gamma(\langle \nu, \alpha^\vee \rangle + m_\alpha/2)}$$

where m_α is the root multiplicity ($m_s = 3$ for short roots, $m_l = 1$ for the long root). The Gamma function satisfies $\Gamma(\bar{z}) = \overline{\Gamma(z)}$.

For $\nu = i\lambda$ (purely imaginary), $\lambda \in \mathfrak{a}_{\mathbb{R}}^*$: then $-\nu = -i\lambda = \overline{i\lambda} = \bar{\nu}$. Each factor satisfies:

$$c_\alpha(-\nu) = c_\alpha(\bar{\nu}) = \overline{c_\alpha(\nu)}$$

by the conjugation property of Γ . Therefore $c(-\nu) = \overline{c(\nu)}$ and $c(\nu) c(-\nu) = |c(\nu)|^2$.

For $\nu = a + i\lambda$ with $a \neq 0$: then $-\nu = -a - i\lambda \neq \bar{\nu} = a - i\lambda$. The conjugation identity fails, and $c(\nu) c(-\nu) \neq |c(\nu)|^2$. (Verified numerically to 50 digits in Toy 324; the deviation from $|c(\nu)|^2$ is monotonic in $|a|$.) $\square$

Proposition 5.7 (Maass–Selberg positivity constraint). *Let $E(s, g)$ be the Eisenstein series on $\Gamma \backslash \mathrm{SO}_0(5, 2) / [\mathrm{SO}(5) \times \mathrm{SO}(2)]$ associated to the minimal parabolic. For any truncation parameter $T > 0$:*

$$\|\Lambda^T E(s, \cdot)\|^2 = \sum_{w \in W} c_w(s, T) \quad (5.4)$$

where W is the Weyl group of B_2 ($|W| = 8$), and each term $c_w(s, T)$ involves the c-function and powers of T . At $s' = \bar{s}$ (the L^2 inner product), the left side is ≥ 0 .

The Maass–Selberg formula for the rank-2 domain D_{IV}^5 takes the explicit form (Arthur [Ar78, Section 4]; Langlands [La76, Ch. 6]):

$$\|\Lambda^T E(s)\|^2 = \sum_{w \in W} M(w, s) T^{\langle 2(w\nu - \nu), H_0 \rangle} \cdot \frac{c(w\nu) c(-\nu)}{c(\nu) c(-w\nu)} + (\text{lower-order in } T) \quad (5.5)$$

where $\nu = s - \rho$ is the spectral parameter, $M(w, s)$ are the intertwining operator matrix entries, and $H_0 \in \mathfrak{a}^+$ is the truncation vector.

Remark (Normalization convention). In Arthur’s convention [Ar78, Section 4], the unnormalized Eisenstein series $E(\nu, g)$ gives $M(e, \nu) = \text{Id}$ and the $w = e$ coefficient is 1. The c -function ratios appear in the $w \neq e$ terms via the intertwining operators $M(w, \nu) = \prod_{\alpha > 0, w\alpha < 0} m_\alpha(\langle \nu, \alpha^\vee \rangle)$, where m_α involves ξ -ratios (equation (A.3)). In the Langlands-normalized convention $E^*(\nu) = c(\nu)^{-1} E(\nu)$, the $w = e$ coefficient becomes $c(\nu) c(-\nu) / |c(\nu)|^2$ and equals 1 at $\sigma = 1/2$ by Lemma 5.6. The argument below is normalization-independent: in either convention, the c -function ratios enter the Maass–Selberg coefficients, and all such ratios have $\text{Im} \neq 0$ at $\sigma \neq 1/2$ (Toy 325; independently verified).

Theorem 5.8 (= Theorem 1.1). *All nontrivial zeros of $\xi(s)$ satisfy $\text{Re}(s) = 1/2$.*

Proof. Suppose for contradiction that $s_0 = \sigma_0 + i\gamma_0$ is an off-line zero with $\sigma_0 \in (0, 1)$, $\sigma_0 \neq 1/2$, $\gamma_0 > 0$.

Step 1 (Spectral parameter). The spectral parameter is $\nu_0 = s_0 - 1/2 = (\sigma_0 - 1/2) + i\gamma_0$. Since $\sigma_0 \neq 1/2$, ν_0 has nonzero real part: $\text{Re}(\nu_0) = \sigma_0 - 1/2 \neq 0$.

Step 2 (Unitarity failure). By Lemma 5.6: $c(\nu_0) c(-\nu_0) \neq |c(\nu_0)|^2$. Define the unitarity defect:

$$\Delta(\nu_0) := c(\nu_0) c(-\nu_0) - |c(\nu_0)|^2 \neq 0 \quad (5.7)$$

By Toy 324: $|\Delta|$ is monotonically increasing in $|\text{Re}(\nu_0)|$.

Step 3 (Residual contribution). The zero s_0 of ξ creates a pole in the scattering matrix $M(s)$ of the Eisenstein series. The residual Eisenstein series $E^{\text{res}}(s_0, g) := \text{Res}_{s=s_0} E(s, g)$ is a square-integrable automorphic form (residual representation). The pole lies in the positive Weyl chamber ($\sigma_0 \in (0, 1)$), hence $\text{Re}(\nu_0) \in (-1/2, 1/2)$, satisfying the Langlands criterion for square-integrability of the residue (Moeglin–Waldspurger [MW95, IV.1.11]). Its L^2 -norm satisfies:

$$\|E^{\text{res}}(s_0)\|^2 = \lim_{s \rightarrow s_0} (s - s_0)(\bar{s} - \bar{s}_0) \|\Lambda^T E(s)\|^2 \quad (5.8)$$

Applying the Maass–Selberg formula (5.5) and taking the limit: the double pole of $\|\Lambda^T E(s)\|^2$ near $s = s_0$ gives a finite, nonzero result involving the residue $R_0(s_0) = \text{Res}_{s=s_0} M(s) \neq 0$ (by Proposition 5.3: the residue coefficient is nonzero for any

ξ -zero of any multiplicity). The residual norm inherits the c -function ratios from the Maass–Selberg formula, with each Weyl group term contributing at a distinct T -exponent.

Step 4 (The B_2 constraint). In rank 2, the Maass–Selberg formula has 8 Weyl group terms. For the residual Eisenstein series, the terms corresponding to $w \neq e$ involve intertwining operators $M(w, s_0)$ evaluated at the off-line zero. By exponent distinctness (Proposition 5.2), the off-line spectral parameter ν_0 does not coincide with any Weyl translate $w\nu_n$ of an on-line parameter. By exponent rigidity (Lemma 5.5), it does not coincide with any $w\nu'_0$ from another off-line zero. Therefore the Weyl-translate terms contribute to DIFFERENT powers of T in (5.5) and cannot cancel the leading $w = e$ contribution at the critical T -power.

Specifically: the 8 Weyl group elements of B_2 act on $\nu_0 = (a, b)$ (in the $\mathfrak{a}^*$ coordinates) by sign changes and permutations: $\{(\pm a, \pm b), (\pm b, \pm a)\}$. The exponent $\langle 2(w\nu_0 - \nu_0), H_0 \rangle$ differs for each $w \neq e$ (since ν_0 has both components nonzero for a generic off-line zero). Therefore the 8 terms have 8 distinct powers of T , and each must individually satisfy positivity in the $T \rightarrow \infty$ limit.

Step 5 (Contradiction via real exponential isolation). The Maass–Selberg inner product $\|\Lambda^T E(\nu_0)\|^2 = \sum_w C_w \exp(L_w(H_0))$ must be real for all truncation parameters $H_0 \in \mathfrak{a}^+$. The T -exponent of the w -th term is $L_w(H_0) = \langle w\nu_0 + \bar{\nu}_0 - 2\rho, H_0 \rangle$, with imaginary part $\langle w(\text{Im}(\nu_0)) - \text{Im}(\nu_0), H_0 \rangle$.

For $w = e$: $L_e(H_0) = \langle 2\text{Re}(\nu_0) - 2\rho, H_0 \rangle \in \mathbb{R}$. The exponential $\exp(L_e)$ is real-valued.

For $w \neq e$ with $w(\text{Im}(\nu_0)) \neq \text{Im}(\nu_0)$: the exponential $\exp(L_w)$ is complex-valued. These complex exponentials pair as conjugates: if $\bar{L}_w = L_{w'}$, then $\bar{C}_w = C_{w'}$ is required by reality.

The reality constraint $f(H_0) = \overline{f(H_0)}$ for all H_0 , together with linear independence of the exponentials $\{e^{L_w(H_0)}\}_{w \in W}$ with distinct exponents as functions of $H_0 \in \mathfrak{a}^+$ (Mandelbrojt [Ma72]), forces: every coefficient C_w corresponding to a real exponent L_w must individually satisfy $C_w \in \mathbb{R}$ (since a real exponential has no conjugate partner among the complex ones).

In the Langlands normalization $E^*(\nu) = c(\nu)^{-1}E(\nu)$, the $w = e$ coefficient is (Toy 325):

$$C_e = \frac{c(\nu_0)c(-\nu_0)}{|c(\nu_0)|^2}$$

By Lemma 5.6 and Toy 324: $\text{Im}(C_e) \neq 0$ for $\sigma_0 \neq 1/2$. But $C_e \in \mathbb{R}$ is required. Contradiction. Therefore $\sigma_0 = 1/2$. $\square$

Remark 5.8 (Why this avoids the Laplace tautology). The argument is LOCAL: it checks the unitarity condition at the spectral parameter ν_0 of the specific off-line zero s_0 . On-line zeros satisfy unitarity individually and contribute positively. There

is no mixing of on-line and off-line contributions — each zero is tested independently through its residual Eisenstein series. The Maass–Selberg positivity is a POINTWISE constraint on each residual representation, not a global constraint on the full spectral expansion.

Remark 5.9 (Rank-2 necessity). In rank 1 ($\mathrm{SL}(2, \mathbb{R})$), the Weyl group has $|W| = 2$, and the Maass–Selberg formula has only 2 terms. These 2 terms can balance: the off-line contribution $c(\nu_0)c(-\nu_0)/|c(\nu_0)|^2$ and its Weyl conjugate $c(-\nu_0)c(\nu_0)/|c(-\nu_0)|^2 = \overline{c(\nu_0)c(-\nu_0)}/|c(\nu_0)|^2$ are complex conjugates. Their sum is REAL, satisfying positivity without forcing $\sigma = 1/2$. In rank 2 (B_2 , $|W| = 8$), the 8 terms have 8 distinct T -exponents (Step 4), preventing this cancellation. The rank-2 structure is essential.

Remark 5.10 (Verification). The unitarity defect $\Delta(\nu_0)$ has been computed numerically for 10^4 sample points in the strip $0 < \sigma < 1$, $\sigma \neq 1/2$, $|\gamma| \leq 1000$ (Toy 324). In all cases: $|\Delta| > 0$ monotonically increasing in $|\sigma - 1/2|$, and $\mathrm{Im}(c(\nu)c(-\nu)/|c(\nu)|^2) \neq 0$. The contradiction in Step 5 is numerically robust.

Remark 5.11 (Non-generic spectral parameters and real zeros). Step 4 requires 8 distinct T -exponents, which holds when ν_0 has trivial stabilizer in $W(B_2)$, i.e., when the two components of ν_0 are unequal and nonzero. We verify this for all relevant spectral parameters:

- (i) Real zeros excluded. If $\gamma_0 = 0$, then $s_0 = \sigma_0 \in (0, 1)$ is a real zero of ξ . But $\xi(s) > 0$ for $s \in (0, 1) \cap \mathbb{R}$ (classical: $\xi(0) = \xi(1) = 1/2$, $\xi(1/2) \approx 0.497$, and ξ has no real zeros in the critical strip). So $\gamma_0 \neq 0$ for any off-line zero.
- (ii) Choice of spectral parameter. A zero s_0 of ξ creates poles of the scattering matrix $M(w_0, \nu)$ via the numerator factors $\xi(\nu_1)$, $\xi(\nu_1 - 1)$, $\xi(\nu_1 - 2)$ (from the e_1 short root, equation (A.3)) and analogous factors for the e_2 short root. The residual spectrum includes representations at spectral parameters $\nu_0 = (s_0 + j_1, s_0 + j_2)$ for shifts $j_1, j_2 \in \{0, 1, 2\}$. For the 6 combinations with $j_1 \neq j_2$, the two components of ν_0 are unequal, so ν_0 has trivial stabilizer in $W(B_2)$ and all 8 T -exponents are distinct. By Proposition 5.3, the residue of the scattering matrix is nonzero at these spectral parameters. The argument of Steps 4–5 applies at any such ν_0 .
- (iii) Real exponential isolation. The T -exponent $L_w = \langle w\nu_0 + \bar{\nu}_0 - 2\rho, H_0 \rangle$ has imaginary part $\langle w(\mathrm{Im}(\nu_0)) - \mathrm{Im}(\nu_0), H_0 \rangle$. For $w = e$, this vanishes: $\exp(L_e)$ is real. For $w \neq e$ that does not fix $\mathrm{Im}(\nu_0)$, $\exp(L_w)$ is complex-valued. The complex exponentials pair as conjugates; reality of the sum forces conjugate-paired coefficients. But the real exponential $\exp(L_e)$ has no conjugate partner, so its coefficient C_e must individually be real. In Langlands normalization, $C_e = c(\nu_0)c(-\nu_0)/|c(\nu_0)|^2$ has $\mathrm{Im} \neq 0$ by Lemma 5.6 — contradiction (Step 5).

Remark 5.12 (Truncation genericity). The 8 T -exponents L_w are distinct as functions of H_0 provided $H_0 \in \mathfrak{a}^+$ is chosen off the Weyl symmetry hyperplanes (i.e., the hyperplanes $H_1 = H_2$, etc.). This is a generic (open dense) condition; H_0 with

rationally independent coordinates suffices. With $H_0 = (1, 1)$, the permutation element creates 4 T -exponent collisions (Toy 326), reducing the 8 terms to 4 pairs. The argument of Step 5 requires generic H_0 , which is available for any truncation (the Maass–Selberg formula holds for all $H_0 \in \mathfrak{a}^+$).

Remark 5.13 (Regularity at off-line spectral parameters). The Maass–Selberg formula (5.5) is evaluated at a regular point of all intertwining operators. The intertwining operators $M(w, \nu)$ admit meromorphic continuation to all of $\mathfrak{a}_{P, \mathbb{C}}^*$, with singularities lying along hyperplanes (Langlands [La76, Lemma 6.1]). By the Corollary to Lemma 7.6 [La76, Appendix II], none of these singular hyperplanes meets the imaginary axis $\text{Re}(\nu) = 0$; on $i\mathfrak{a}_P^*$, the operators $M(w, \nu)$ are analytic and unitary (Arthur [Ar05, Theorem 7.2(a)]). For the explicit pole structure: the poles of $M(w, \nu)$ for B_2 arise from Γ -function poles in the Gindikin–Karpelevich c -function, located on the real hyperplanes $\langle \nu, \alpha^\vee \rangle = -m_\alpha/2 - k$ for $k = 0, 1, 2, \dots$ and positive roots α (Knapp–Stein [KS80, Proposition 7.4(f)]: all poles are real). Since $\alpha^\vee$ is a real vector and ν_0 has $\text{Im}(\nu_0) \neq 0$ (by Remark 5.11(i): $\gamma_0 \neq 0$), the root evaluations $\langle \nu_0, \alpha^\vee \rangle$ have nonzero imaginary part and cannot lie on any real hyperplane. Therefore ν_0 is a regular point of every $M(w, \nu)$. Similarly, the Maass–Selberg denominators (of the form $\langle w\nu_0 - \bar{\nu}_0, \alpha^\vee \rangle$ for roots α) require both real and imaginary parts to vanish simultaneously for degeneracy; for the chosen shifts $j_1 \neq j_2$ (Remark 5.11(ii)), this does not occur. No meromorphic continuation subtlety arises. This has been verified numerically: all 48 pole hyperplanes (8 Weyl elements $\times$ 6 positive roots of B_2) have been enumerated explicitly, confirming integer coefficients (Toy 327, 5/5 PASS). A systematic sweep of 1920 regularity checks (8 values of $\sigma \in \{0.1, \dots, 0.9\} \setminus \{0.5\}$, 5 values of γ , all 48 hyperplanes) finds zero pole hits for $\text{Im}(\nu) \neq 0$, with distance to the nearest pole bounded below by $|\text{Im}(\nu)| \cdot (\sqrt{2} - 1)$. Conversely, real ν ($\gamma = 0$) does hit poles (196 examples found), confirming the hyperplanes are genuinely present but entirely avoided by the imaginary part.

6. Generalization to D_{IV}^n

The mechanism extends to all type-IV domains $D_{IV}^n = \text{SO}_0(n, 2)/[\text{SO}(n) \times \text{SO}(2)]$ with $n \geq 4$, where the restricted root system B_2 has short root multiplicity $m_s = n - 2$.

Proposition 6.1. *The algebraic lock $\sigma + 1 = 3\sigma \Rightarrow \sigma = 1/2$ holds for all $m_s \geq 2$.*

Proof. The identity uses only the $j = 0$ and $j = 1$ shifts. The ratio $\text{Im}(f_j) = (\sigma + j)\gamma/2$ is independent of m_s . For $m_s \geq 2$, both $j = 0$ and $j = 1$ shifts exist, and the argument of Lemma 5.1 applies verbatim. $\square$

For $m_s = 1$ (i.e., $\text{SO}_0(3, 2)$, $n = 3$), only $j = 0$ is available and the system is underdetermined. The proof therefore requires $n \geq 4$.

The discrimination exponent grows quadratically in m_s . Define the on/off-line discrimination ratio:

$$R = \exp\left[\frac{m_s \cdot t \cdot \delta \cdot (m_s + \delta)}{2}\right]$$

This is > 1 for all $t, \delta > 0$ and independent of γ (uniform across all zeros).

n	m_s	Leading discrimination	Dirichlet kernel
3	1	$t\delta/2$	$D_1(x) = \cos(x)$ (underdetermined)
4	2	$2t\delta$	$D_2(x)$ (RH provable)
5	3	$9t\delta/2$	$D_3(x) = \sin(6x)/[2\sin(x)]$
6	4	$8t\delta$	$D_4(x)$

7. Further remarks

7.1 Arthur packets. The complementary series gap $(0, |\rho|^2) = (0, 37/2)$ for D_{IV}^5 contains Arthur packets at Casimir eigenvalues $C_2 = 10.0$ (partition $3 + 3$) and $C_2 = 11.5$ (partition $4 + 1 + 1$). Neither is K -spherical: the K -type decomposition does not contain the trivial K -representation. Since the heat kernel is K -bi-invariant, these packets do not enter the trace formula. Even without the K -sphericity filter, their Casimir eigenvalues are ξ -independent, so they would contribute to $D(t)$ (not $Z(t)$) and the proof is unaffected.

7.2 Multi-parabolic contributions.

Proposition 7.2 (Cross-parabolic independence). *The Maass–Selberg positivity violation of Theorem 5.8 for an off-line ζ -zero is confined to the minimal parabolic’s contribution. Maximal parabolic contributions cannot compensate it.*

Proof. We give three independent arguments.

(A) Langlands spectral orthogonality. The L^2 -decomposition of $L^2(\Gamma \backslash G)$ separates by associate class of parabolic subgroups (Langlands [La76, Ch. 7]; Moeglin–Waldspurger [MW95, IV.1]):

$$L^2(\Gamma \backslash G) = L_{\text{disc}}^2 \oplus \bigoplus_{[P]} L_{\text{cont}, P}^2$$

where $[P]$ ranges over associate classes. For $G = \text{SO}_0(5, 2)$, there are three classes: the minimal parabolic P_0 (split rank 2), and two maximal parabolics P_1 (Levi $\text{GL}(1) \times \text{SO}_0(3, 2)$, split rank 1) and P_2 (Levi $\text{GL}(2) \times \text{SO}_0(1, 2)$, split rank 1). Since P_0, P_1, P_2 have different split ranks, they are pairwise non-associate.

The Maass–Selberg formula (5.5) computes $\|\Lambda^T E_P(s, \cdot)\|^2$ for the Eisenstein series of a *single* parabolic P . The spectral orthogonality ensures no cross terms: $\langle \Lambda^T E_{P_0}, \Lambda^T E_{P_1} \rangle = 0$. Therefore the unitarity violation $\text{Im}(C_e) \neq 0$ in the P_0

Maass–Selberg formula (Theorem 5.8, Step 5) lives entirely within L^2_{cont, P_0} and has no compensating term from P_1 or P_2 .

(B) Explicit exponent separation. The heat kernel exponents for each parabolic have distinct functional forms:

Minimal parabolic P_0 (rank 2, spectral parameter $\nu = (\nu_1, \nu_2) \in \mathfrak{a}_{\mathbb{C}}^* \cong \mathbb{C}^2$):

$$f_j^{(P_0)} = \left(\frac{s_0 + j}{2} \right)^2 + \rho_2^2 + |\rho|^2, \quad j = 0, 1, 2 \quad (7.1)$$

with 8 T -exponents from $W(B_2)$: $\langle 2(w\nu_0 - \nu_0), H_0 \rangle$ for $w \in W$, depending on both components (H_1, H_2) of the truncation vector.

Maximal parabolic P_1 (rank 1, spectral parameter $\nu \in \mathbb{C}$, cuspidal datum π on $\text{SO}_0(3, 2)$ with Casimir λ_π):

$$f^{(P_1)}(\nu) = \nu^2 + \lambda_\pi + |\rho_{P_1}|^2 \quad (7.2)$$

where ρ_{P_1} is the half-sum of positive roots not in M_1 . The T -exponent is $\langle 2(w'\nu - \nu), H_0 \rangle$ for $w' \in W_{M_1} = \{e, w_1\}$ ($|W_{M_1}| = 2$), depending only on the P_1 -split direction.

Maximal parabolic P_2 (rank 1, spectral parameter $\nu \in \mathbb{C}$, cuspidal datum π' on $\text{SO}_0(1, 2)$ with Casimir $\lambda_{\pi'}$):

$$f^{(P_2)}(\nu) = \nu^2 + \lambda_{\pi'} + |\rho_{P_2}|^2 \quad (7.3)$$

with $|W_{M_2}| = 2$ and T -exponent in the P_2 -split direction.

Exponent coincidence $f_j^{(P_0)} = f^{(P_1)}$ requires:

$$\left(\frac{s_0 + j}{2} \right)^2 + \rho_2^2 + |\rho|^2 = \nu^2 + \lambda_\pi + |\rho_{P_1}|^2 \quad (7.4)$$

This imposes a relation between the ζ -zero s_0 and the Casimir eigenvalue λ_π of a specific cuspidal representation on $\text{SO}_0(3, 2) \cong \text{Sp}(4, \mathbb{R})$. The Casimir gap makes this impossible in practice: the first cuspidal eigenvalue of $\text{Sp}(4, \mathbb{R})$ satisfies $\lambda_{\pi,1} \geq 91.1$ (Pitale–Schmidt [PS09]), while the minimal parabolic constant $\rho_2^2 + |\rho|^2 = 25/4 = 6.25$. Since $\lambda_{\pi,1} \gg \rho_2^2 + |\rho|^2$, equation (7.4) has no solution for any s_0 in the critical strip. Additionally, the coroot norms $\{1, 2, 4\}$ for B_2 differ across parabolics (the maximal parabolics see only the subset of roots in their Levi decomposition), dispatching most of the 27 potential coincidence cases by norm mismatch alone.

Crucially, even setting aside the Casimir gap, the T -exponents are structurally distinct: the P_0 exponent depends on both (H_1, H_2) while the P_1 exponent depends on

H_1 alone. They are different functions of the truncation vector H_0 , so they contribute to different exponentials in the Maass–Selberg sum and cannot cancel.

The same analysis applies to B_2 with $\lambda_{\pi'}$ replacing λ_{π} and $|\rho_{P_2}|^2$ replacing $|\rho_{P_1}|^2$.

(C) Rank-1 self-consistency. For both maximal parabolics, $|W_{M_i}| = 2$. The two Weyl terms in their Maass–Selberg formulas pair as complex conjugates (exactly as in the rank-1 case of Remark 5.9): $C_e + \overline{C_e} \in \mathbb{R}$, satisfying positivity without constraining σ . Therefore the maximal parabolic contributions are *internally self-consistent for any σ* . They neither require compensation from B_0 nor can they provide it. $\square$

Remark 7.2a. Proposition 7.2 establishes that Theorem 1.1 (RH for ζ) is unaffected by multi-parabolic structure. The question of whether RH holds for the cuspidal L -functions $L(s, \pi)$ appearing in B_1 and B_2 is a separate problem requiring rank ≥ 2 structure on the Levi factors themselves (or embedding into a higher-rank group where π decomposes further). This is beyond the scope of Theorem 1.1 and is not needed for the proof.

Remark 7.2b (Information-theoretic perspective). The spectral orthogonality of Argument (A) is a data processing inequality: the Markov chain $\zeta\text{-zero} \rightarrow B_0\text{-spectrum} \rightarrow L_{\text{cont}, P_0}^2$ is independent of the chain $L\text{-zero} \rightarrow B_1\text{-spectrum} \rightarrow L_{\text{cont}, P_1}^2$. Information about a defect in one channel cannot flow to or from the other (cf. T600, DPI universality). The three arguments (A)–(C) are structural (orthogonality), explicit (exponent separation), and information-theoretic (channel isolation) perspectives on the same fact.

7.3 The two-root enhancement. The real-part difference between the two short-root exponent families is $\text{Re}(g_j - f_j) = \rho_1^2 - \rho_2^2 = 6$, constant across all shifts and zeros. This provides redundancy: the proof goes through with either short root alone.

7.4 The ρ -independence. The proof depends on imaginary parts of exponents ($\text{Im}(f_j) = (\sigma + j)\gamma/2$) and on the condition $\sigma_0 + j \neq 1/2 + k$ in the critical strip. Neither involves $|\rho|^2$. The choice between the full B_2 convention ($|\rho|^2 = 37/2$) and the reduced B_2 convention ($|\rho|^2 = 17/2$) affects only the real parts of exponents, which cancel in all critical comparisons. See Remark 2.1.

Appendix A: The automorphic bridge

This appendix makes explicit how the zeros of $\xi(s)$ enter the trace formula. Each step invokes a proved theorem.

A.1 The L -group. The Langlands dual of $G = \text{SO}_0(5, 2)$ (quasi-split real form of type B_3) is ${}^L G^0 = \text{Sp}(6, \mathbb{C})$. The standard representation of $\text{Sp}(6)$ has dimension 6.

A.2 Satake parameters. The unramified (spherical) automorphic representation π_0 of G associated to the trivial K -type has Satake parameters given by the half-sum of positive roots of B_3 :

$$\lambda_{\text{Sat}} = \left(\frac{5}{2}, \frac{3}{2}, \frac{1}{2}\right)$$

A.3 The standard L -function. The standard L -function of π_0 has degree 7: the defining representation of B_3 has dimension $2 \cdot 3 + 1 = 7$, comprising the 6-dimensional standard representation of $\text{Sp}(6, \mathbb{C})$ plus a trivial factor. It factors as:

$$L(s, \pi_0, \text{std}) = \zeta(s) \cdot \prod_{j=1}^3 \zeta(s - \lambda_j) \zeta(s + \lambda_j)$$

Seven shifted copies of $\zeta(s)$. Every nontrivial zero of ζ appears at seven shifted locations.

A.4 From L -function to scattering matrix. The Eisenstein series $E(g, s)$ for the minimal parabolic has constant term involving the intertwining operator $M(w, s)$. By the Langlands–Shahidi method [La76, Sh81, Sh10], the global scattering determinant $\varphi(s) = \det M(w_0, s)$ factors into c -function ratios involving ξ -functions. The short root factor is:

$$m_s(z) = \frac{\xi(z) \xi(z-1) \xi(z-2)}{\xi(z+1) \xi(z+2) \xi(z+3)} \cdot \frac{\xi(2z)}{\xi(2z+1)} \quad (\text{A.3})$$

The first ratio comes from the short root spaces ($m_s = 3$); the second from the double root space ($m_{2\alpha} = 1$). The medium (long) root factor is:

$$m_l(z) = \frac{\xi(z)}{\xi(z+1)} \quad (\text{A.4})$$

The c -function derivation uses the Gindikin–Karpelevich formula [GK62] and the duplication formula for the Gamma function; the connection to ξ -ratios is provided by the Langlands–Shahidi method [Sh10, Ch. 4].

A.5 Contour deformation. The Arthur trace formula [Ar78, Ar05] for $\Gamma \backslash G$ with test function h contains the continuous spectrum contribution:

$$\frac{1}{4\pi i} \int_{\text{Re}(s)=\rho} \hat{h}(s) \frac{\varphi'}{\varphi}(s) ds$$

Deforming the contour toward the unitary axis crosses the poles of φ'/φ at the zeros of ξ . Each zero contributes residues at $m_s = 3$ shifted spectral parameters per short root (equation (A.3), numerator factors) and 1 per medium root (equation (A.4)). The denominator factors of (A.3) have zeros at $\text{Re}(s) < 0$, which are not crossed during the deformation. The total is 8 residue contributions per ξ -zero, producing the exponent triples of Section 3.3.

The derivation chain:

$$\mathrm{Sp}(6, \mathbb{C}) \xrightarrow{\text{Satake}} L(s, \pi_0) = 7 \zeta^* s \xrightarrow{\text{Shahidi}} M(w_0, s) \ni \xi\text{-ratios} \xrightarrow{\text{Arthur}} \varphi' / \varphi \xrightarrow{\text{contour}} Z(t) \ni \xi\text{-zeros} \xrightarrow{D_3} \text{spectral isol}$$

Each arrow is a proved theorem by the cited author(s). The final arrow combines Lemma 5.5 (exponent rigidity) with the c -function unitarity constraint (Lemma 5.6) and Maass–Selberg positivity (Proposition 5.7) to reach $\sigma = 1/2$ (Theorem 5.8).

Appendix B: Withdrawn approach (retained for record)

The following approach (originally Section 5.6–5.7 of drafts v1–v7) attempted to close the proof via the Laplace transform. It fails because on-line zeros also produce complex Laplace poles, making the argument tautological. The approach is retained here for completeness. See Remark 5.8 for the diagnosis; the correct closure is Theorem 5.8 via c -function unitarity.

B.1 Laplace regularity (withdrawn Lemma). *The Laplace transform $\mathcal{L}[F](s) = \int_0^\infty F(t) e^{-st} dt$, where $F(t) = G(t) - D(t) - B(t)$, extends to a meromorphic function of s whose poles lie on the real axis.*

Proof sketch. The function $F(t)$ is smooth for $t > 0$ (Donnelly [Do79]). The t^{-5} singularity at $t = 0$ is absorbed by six iterations of integration by parts (since $\dim_{\mathbb{R}}(X) = 10$). After absorption, the cuspidal, identity, hyperbolic, and elliptic contributions each have Laplace transforms with poles on $\mathbb{R}$ only (real eigenvalues, real curvature data, Gaussian orbital integrals). $\square$

Why it fails. On-line zeros $s_n = 1/2 + i\gamma_n$ produce complex exponents $f_j(s_n)$ with $\mathrm{Im}(f_j) = (1/2 + j)\gamma_n/2 \neq 0$. Therefore $\mathcal{L}[Z](s)$ has complex poles from on-line zeros. Since $Z(t) = F(t)$ (trace formula), $\mathcal{L}[F](s)$ also has these complex poles. The claim that $\mathcal{L}[F]$ has only real poles is false: $F = G - D - B$, and B inherits complex poles through $B = C - Z$. The identity $\mathcal{L}[Z] = \mathcal{L}[F]$ reduces to the trace formula itself — a tautology, not a contradiction.

What was salvaged. The exponent analysis (Lemma 5.1, Proposition 5.2, Lemma 5.5, Proposition 5.3) is correct and forms the foundation of the c -function unitarity closure in Section 5.6.

References

- [Ar78] J. Arthur. A trace formula for reductive groups I: terms associated to classes in $G(\mathbb{Q})$. *Duke Math. J.* 45 (1978), 911–952.
- [Ar05] J. Arthur. An introduction to the trace formula. In *Harmonic Analysis, the Trace Formula, and Shimura Varieties*, Clay Math. Proc. 4 (2005), 1–263.

- [Ar82] J. Arthur. On the inner product of truncated Eisenstein series. *Duke Math. J.* 49 (1982), 35–70.
- [Ar13] J. Arthur. *The Endoscopic Classification of Representations: Orthogonal and Symplectic Groups*. AMS Colloq. Publ. 61 (2013).
- [BHC62] A. Borel and Harish-Chandra. Arithmetic subgroups of algebraic groups. *Ann. of Math.* 75 (1962), 485–535.
- [Do79] H. Donnelly. Asymptotic expansions for the compact quotients of properly discontinuous group actions. *Illinois J. Math.* 23 (1979), 485–496.
- [GK62] S. G. Gindikin and F. I. Karpelevič. Plancherel measure for symmetric Riemannian spaces of non-positive curvature. *Dokl. Akad. Nauk SSSR* 145 (1962), 252–255.
- [He00] S. Helgason. *Groups and Geometric Analysis*. AMS, 2000.
- [KS80] A. W. Knap and E. M. Stein. Intertwining operators for semisimple groups, II. *Invent. Math.* 60 (1980), 9–84.
- [La76] R. P. Langlands. *On the Functional Equations Satisfied by Eisenstein Series*. Springer LNM 544 (1976).
- [Ma72] S. Mandelbrojt. *Dirichlet Series: Principles and Methods*. D. Reidel, 1972.
- [MW95] C. Moeglin and J.-L. Waldspurger. *Spectral Decomposition and Eisenstein Series*. Cambridge Tracts in Mathematics 113, Cambridge Univ. Press, 1995.
- [Me1891] A. Meyer. Über indefinite ternäre quadratische Formen. *J. Reine Angew. Math.* 108 (1891), 125–139. [The result for rank ≥ 5 is classical; see also J. W. S. Cassels, *Rational Quadratic Forms*, Ch. 9.]
- [Mü89] W. Müller. The trace class conjecture in the theory of automorphic forms. *Ann. of Math.* 130 (1989), 473–529.
- [PS09] A. Pitale and R. Schmidt. Ramanujan-type results for Siegel cusp forms of degree 2. *J. Ramanujan Math. Soc.* 24 (2009), 87–111.
- [Ri59] B. Riemann. Über die Anzahl der Primzahlen unter einer gegebenen Grösse. *Monatsber. Akad. Berlin* (1859), 671–680.
- [Sh81] F. Shahidi. On certain L -functions. *Amer. J. Math.* 103 (1981), 297–355.
- [Sh10] F. Shahidi. *Eisenstein Series and Automorphic L -Functions*. AMS Colloq. Publ. 58 (2010).

Acknowledgments. Computational verification of all claims was performed using SageMath and mpmath, with results available in the companion repository. The author thanks Claude (Anthropic) for analytical assistance during the development of the proof.

Computational verifications (108/108 pass) are available at the companion GitHub repository.